NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



AI READY

AI Laptop Selection Guide

CPU, GPU, NPU, RAM and VRAM explained for AI builders

NexaStore Knowledge Series • 2026 Edition

1. There Is No Single 'AI Laptop'

AI workloads range from efficient on-device features to GPU-intensive local LLM inference and computer vision. Choose the accelerator according to the software you intend to run. CPU, GPU, NPU, system RAM and storage each solve different constraints.

AI workload	Most important hardware	Why
Cloud LLM application	CPU, RAM, battery	Inference runs remotely; local machine hosts development tools and services.
On-device productivity AI	NPU, efficient CPU	NPU can accelerate supported low-power local AI experiences.
Local LLM inference	GPU VRAM + system RAM	Model weights and runtime state must fit available memory.
Computer vision	Discrete GPU	Image models benefit strongly from parallel GPU compute.
RAG development	RAM, CPU, SSD; optional GPU	Ingestion, databases, containers and embeddings may run locally.
Fine-tuning experiments	High-VRAM GPU	Training memory requirements can exceed inference requirements.

2. VRAM Tiers

GPU VRAM	Typical educational use	Limitations
Integrated / shared	Cloud-first AI, lightweight local inference	Uses system memory; software/runtime support varies
8 GB	Small quantized models, embeddings, CV prototyping	Limited headroom for larger models/context
12 GB	Stronger local inference and development	Still constrained for larger models
16 GB	Advanced local AI development	Better model/context headroom; higher cost and power
24 GB+	High-end mobile/pro workstation class	Premium, heavier systems; desktop/cloud may offer better value

3. NPU vs GPU

Characteristic	NPU	Discrete GPU
Primary strength	Efficient supported on-device AI	High-throughput parallel compute
Power efficiency	High	Lower under heavy load
Local LLM ecosystem	Growing; runtime dependent	Broad for many established GPU runtimes
Computer vision / ML dev	Workload dependent	Strong
Battery-oriented AI	Excellent fit	Less ideal for sustained use on battery
Replaces the other?	No	No

4. Suggested AI Laptop Profiles

Profile	RAM	GPU VRAM	Typical user
AI-enabled developer	32 GB	Optional / 8 GB	Cloud AI, RAG, agents, occasional local models
Local AI developer	64 GB	12-16 GB	Frequent local inference, CV, embeddings
Mobile AI power user	64-128 GB	16-24 GB	Heavier local experimentation and creator/AI mix
Cloud-first enterprise user	32 GB	Optional	Mostly managed AI services, containers and IDEs

5. Recommendation Examples

User request	Reasoning direction
'I build .NET agents with Azure OpenAI.'	Prioritize CPU, 32 GB+ RAM, battery and developer ergonomics; discrete GPU is optional.
'I run Ollama models every day.'	Prioritize system RAM and GPU VRAM; verify runtime/GPU compatibility.
'I work on computer vision and CUDA.'	Prioritize compatible NVIDIA-class discrete GPU and sufficient VRAM.
'I need AI features but travel constantly.'	Favor efficient CPU/NPU, battery life and weight; use cloud GPU for heavy workloads.
'I run Docker, Kubernetes, PostgreSQL and local AI together.'	Favor 64 GB RAM, fast 2 TB SSD and discrete GPU if local inference is frequent.

6. RAG and Commerce Data

This guide explains suitability and trade-offs. Exact SKU specifications, price, stock and current configuration must be retrieved from structured NexaStore product and inventory services. An AI recommendation should combine those live facts with knowledge retrieved from this guide.

Educational notice

This document belongs to the fictional NexaStore educational case study.